

Question 2.1: Why does the Average Variable Cost (AVC) not change when going from output level 5 to output level 6?

Type your answer here, replacing this text.

0.0.1 Question 4

Why does the Marginal Cost line intersect with the Average Variable Cost and Average Total Cost at their minima?

Type your answer here, replacing this text.

Question 6.10: Create your own scenario of a shift in the supply curve or the demand curve in a market of your choice. Please include the market, what caused the shift, which direction it shifts supply/demand, and what happens to the new equilibrium price and quantity.

Type your answer here, replacing this text.

Congratulations! You just finished Lab 2! Feel free to use the box below to let us know if you have any feedback!

How did you find this lab? Any improvements? Any questions you particularly enjoyed working through?

0.1 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a zip file for you to submit. **Please save before exporting!**

```
In [11]: # Save your notebook first, then run this cell to export your submission.
         grader.export(run_tests=True)
```

Running your submission against local test cases...

Your submission received the following results when run against available test cases:

q1_1 results: All test cases passed!

q1_2 results: All test cases passed!

q1_3 results:

q1_3 - 1 result:

Test case failed

Trying:

type(iii) == int

Expecting:

True

Line 1, in q1_3 0

Failed example:

type(iii) == int

Expected:

True

Got:

False

q1_4 results:

```

q1_4 - 1 result:
    Test case failed
    Trying:
        type(iv) == int
    Expecting:
        True
    *****
    Line 1, in q1_4 0
    Failed example:
        type(iv) == int
    Expected:
        True
    Got:
        False

q1_5 results:
    q1_5 - 1 result:
        Test case failed
        Trying:
            type(v) == int
        Expecting:
            True
        *****
        Line 1, in q1_5 0
        Failed example:
            type(v) == int
        Expected:
            True
        Got:
            False

q1_6 results:
    q1_6 - 1 result:
        Test case failed
        Trying:
            type(vi) == int
        Expecting:
            True
        *****
        Line 1, in q1_6 0
        Failed example:
            type(vi) == int
        Expected:
            True
        Got:
            False

q1_7 results:
    q1_7 - 1 result:
        Test case failed
        Trying:
            type(vii) == int
        Expecting:
            True

```

```

*****
Line 1, in q1_7 0
Failed example:
    type(vii) == int
Expected:
    True
Got:
    False

q1_8 results:
  q1_8 - 1 result:
    Test case failed
    Trying:
        type(viii) == int
    Expecting:
        True
*****
Line 1, in q1_8 0
Failed example:
    type(viii) == int
Expected:
    True
Got:
    False

q2_2 results:
  q2_2 - 1 result:
    Test case failed
    Trying:
        assert firm_will_produce in [True, False]
    Expecting nothing
*****
Line 1, in q2_2 0
Failed example:
    assert firm_will_produce in [True, False]
Exception raised:
    Traceback (most recent call last):
      File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
        exec(compile(example.source, filename, "single",
      File "<doctest q2_2 0[0]>", line 1, in <module>
        assert firm_will_produce in [True, False]
        ~~~~~~
    AssertionError
    Trying:
        assert output >= 0
    Expecting nothing
*****
Line 2, in q2_2 0
Failed example:
    assert output >= 0
Exception raised:
    Traceback (most recent call last):
      File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
        exec(compile(example.source, filename, "single",

```

```

File "<doctest q2_2 0[1]>", line 1, in <module>
    assert output >= 0
    ~~~~~
TypeError: '>=' not supported between instances of 'ellipsis' and 'int'

q3 results:
q3 - 1 result:
    Test case failed
    Trying:
        assert q3 in ['Yes', 'No']
    Expecting nothing
    *****
    Line 1, in q3 0
    Failed example:
        assert q3 in ['Yes', 'No']
    Exception raised:
    Traceback (most recent call last):
      File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
        exec(compile(example.source, filename, "single",
      File "<doctest q3 0[0]>", line 1, in <module>
        assert q3 in ['Yes', 'No']
        ~~~~~
    AssertionError

q5 results:
q5 - 1 result:
    Test case failed
    Trying:
        assert AFC in ['decreases', 'increases', 'stays same'] and ATC_minus_AVC in ['decreases
    Expecting nothing
    *****
    Line 1, in q5 0
    Failed example:
        assert AFC in ['decreases', 'increases', 'stays same'] and ATC_minus_AVC in ['decreases
    Exception raised:
    Traceback (most recent call last):
      File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
        exec(compile(example.source, filename, "single",
      File "<doctest q5 0[0]>", line 1, in <module>
        assert AFC in ['decreases', 'increases', 'stays same'] and ATC_minus_AVC in ['decreases
        ~~~~~
    AssertionError

q6_1 results:
q6_1 - 1 result:
    Test case failed
    Trying:
        str(P_S) == 'P_S'
    Expecting:
        True
    *****
    Line 1, in q6_1 0
    Failed example:
        str(P_S) == 'P_S'

```



```

Expected:
    True
Got:
    False

q6_1 - 2 result:
    Test case failed
Trying:
    str(Q_S) == 'Q_S'
Expecting:
    True
*****
Line 1, in q6_1 1
Failed example:
    str(Q_S) == 'Q_S'
Expected:
    True
Got:
    False

q6_1 - 3 result:
    Test case failed
Trying:
    type(P_S) == sympy.Symbol
Expecting:
    True
*****
Line 1, in q6_1 2
Failed example:
    type(P_S) == sympy.Symbol
Expected:
    True
Got:
    False

q6_1 - 4 result:
    Test case failed
Trying:
    type(Q_S) == sympy.Symbol
Expecting:
    True
*****
Line 1, in q6_1 3
Failed example:
    type(Q_S) == sympy.Symbol
Expected:
    True
Got:
    False

q6_2 results:
    q6_2 - 1 result:
        Test case failed
        Trying:

```

```

len(demand_curve_properties) == 2
Expecting:
True
*****
Line 1, in q6_2 0
Failed example:
len(demand_curve_properties) == 2
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_2 0[0]>", line 1, in <module>
    len(demand_curve_properties) == 2
    ~~~~~
TypeError: object of type 'ellipsis' has no len()

q6_3 results:
q6_3 - 1 result:
Test case failed
Trying:
assert type(P) == sympy.Symbol
Expecting nothing
*****
Line 1, in q6_3 0
Failed example:
assert type(P) == sympy.Symbol
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_3 0[0]>", line 1, in <module>
    assert type(P) == sympy.Symbol
    ~~~~~
AssertionError
Trying:
assert str(P) == 'P'
Expecting nothing
*****
Line 2, in q6_3 0
Failed example:
assert str(P) == 'P'
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_3 0[1]>", line 1, in <module>
    assert str(P) == 'P'
    ~~~~~
AssertionError

q6_3 - 2 result:
Test case failed
Trying:
assert len(demand.as_coefficients_dict()) == 2

```

```

Expecting nothing
*****
Line 1, in q6_3 1
Failed example:
    assert len(demand.as_coefficients_dict()) == 2
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_3 1[0]>", line 1, in <module>
    assert len(demand.as_coefficients_dict()) == 2
           ~~~~~~
NameError: name 'demand' is not defined
Trying:
    assert P in demand.as_coefficients_dict()
Expecting nothing
*****
Line 2, in q6_3 1
Failed example:
    assert P in demand.as_coefficients_dict()
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_3 1[1]>", line 1, in <module>
    assert P in demand.as_coefficients_dict()
           ~~~~~~
NameError: name 'demand' is not defined
Trying:
    assert 1 in demand.as_coefficients_dict()
Expecting nothing
*****
Line 3, in q6_3 1
Failed example:
    assert 1 in demand.as_coefficients_dict()
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_3 1[2]>", line 1, in <module>
    assert 1 in demand.as_coefficients_dict()
           ~~~~~~
NameError: name 'demand' is not defined

```

q6_4 results:

q6_4 - 1 result:

Test case failed

Trying:

```
assert q6_4 in ['a', 'b', 'c', 'd']
```

Expecting nothing

Line 1, in q6_4 0

Failed example:

```
assert q6_4 in ['a', 'b', 'c', 'd']
```

```

Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_4 0[0]>", line 1, in <module>
    assert q6_4 in ['a', 'b', 'c', 'd']
    ~~~~~
AssertionError

```

q6_5 results:

```

q6_5 - 1 result:
  Test case failed
  Trying:
    assert P in supply.as_coefficients_dict()
  Expecting nothing
  *****
  Line 1, in q6_5 0
  Failed example:
    assert P in supply.as_coefficients_dict()
  Exception raised:
  Traceback (most recent call last):
    File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
      exec(compile(example.source, filename, "single",
    File "<doctest q6_5 0[0]>", line 1, in <module>
      assert P in supply.as_coefficients_dict()
      ~~~~~
  AttributeError: 'ellipsis' object has no attribute 'as_coefficients_dict'

```

q6_6 results:

```

q6_6 - 1 result:
  Test case failed
  Trying:
    assert q6_6 in ['a', 'b', 'c', 'd']
  Expecting nothing
  *****
  Line 1, in q6_6 0
  Failed example:
    assert q6_6 in ['a', 'b', 'c', 'd']
  Exception raised:
  Traceback (most recent call last):
    File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
      exec(compile(example.source, filename, "single",
    File "<doctest q6_6 0[0]>", line 1, in <module>
      assert q6_6 in ['a', 'b', 'c', 'd']
      ~~~~~
  AssertionError

```

q6_7 results:

```

q6_7 - 1 result:
  Test case failed
  Trying:
    0 <= P_star <= 1000
  Expecting:
    True

```

```

*****
Line 1, in q6_7 0
Failed example:
    0 <= P_star <= 1000
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_7 0[0]>", line 1, in <module>
    0 <= P_star <= 1000
TypeError: '<=' not supported between instances of 'int' and 'ellipsis'

q6_8 results:
  q6_8 - 1 result:
    Test case failed
    Trying:
      15 <= Q_star <= 50
    Expecting:
      True
*****
Line 1, in q6_8 0
Failed example:
    15 <= Q_star <= 50
Exception raised:
Traceback (most recent call last):
  File "/srv/conda/lib/python3.11/doctest.py", line 1355, in __run
    exec(compile(example.source, filename, "single",
  File "<doctest q6_8 0[0]>", line 1, in <module>
    15 <= Q_star <= 50
TypeError: '<=' not supported between instances of 'int' and 'ellipsis'

q6_9 results: All test cases passed!

```

<IPython.core.display.HTML object>